

MA329 Statistical Linear Models

Prof. Jian Qing Shi(史建清)

Email: shjq@sustech.edu.cn

Room 309, Business School Building

TAs: 杜思敏 [email: 12331108@mail.sustech.edu.cn]
惠金燕 [email: 12532238@mail.sustech.edu.cn]
余成耀 [email: 12532239@mail.sustech.edu.cn]
肖涵 [email: 12432282@mail.sustech.edu.cn]

A little bit about me



A little bit about me



- Lectures/Tutorials

- 16:20 – 18:10 Monday: Room 111, Teaching Building 1
- 16:20 – 18:10 Wednesday (odd weeks only): Room 111, Teaching Building 1

- Assessment

- Assignments: about 6 sets, 30%
- Midterm test: 30%
- Final Exam: 40%
- **ONE** A-4 size formula sheet (written on both sides) is allowed in both midterm and final exams.

Resources

- Rencher, AC. and Schaalje, GB (2007) Linear Models in Statistics, Wiley
- Mendenhall W and Sincich T. (2021) A second Course in Statistics: Regression Analysis (8th edition), Pearson/Prentice Hall
- Teaching materials will be available in **BlackBoard**

Approximate Class Agenda

- Ch1. Introduction
- Ch2. Simple linear regression: model, estimation and testing
- Ch3. Matrix algebra
- Ch4. Random vector and multivariate normal distribution
- Ch5. Quadratic forms
- Ch6. Multiple regression
- **Midterm test** (Nov. 5)
- Ch7. Hypothesis testing and confidence intervals
- Ch8. Diagnostics and model building
- Ch9. Analysis of variance models
- **Final exam**

Ch1. Introduction: What is a regression model

Recall: (sample) covariance

sample covariance

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- $\text{Cov}(x, y) > 0 \rightarrow x$ and y are positively correlated
- $\text{Cov}(x, y) < 0 \rightarrow x$ and y are inversely correlated
- $\text{Cov}(x, y) = 0 \rightarrow x$ and y are uncorrelated.

Remark 1.1 * To study the relationship between two variables (r.v.)
 y - son's height x - father's height

$$\begin{aligned}
 * \text{Cov}(X, Y) &= \mathbb{E}(X - \mathbb{E}X)(Y - \mathbb{E}Y) \\
 &= \int (x - \mu_x)(y - \mu_y) p(x, y) dx dy
 \end{aligned}$$

* sample variance based on a dataset
 $\{(x_i, y_i), i = 1, \dots, n\}$

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

large sample theory

$$\xrightarrow{\quad} \text{Cov}(X, Y) \quad \text{as } n \rightarrow \infty$$

Introduction: Correlation Coefficient

Pearson's Correlation Coefficient is standardized covariance
(unitless):

*linear
correlation*

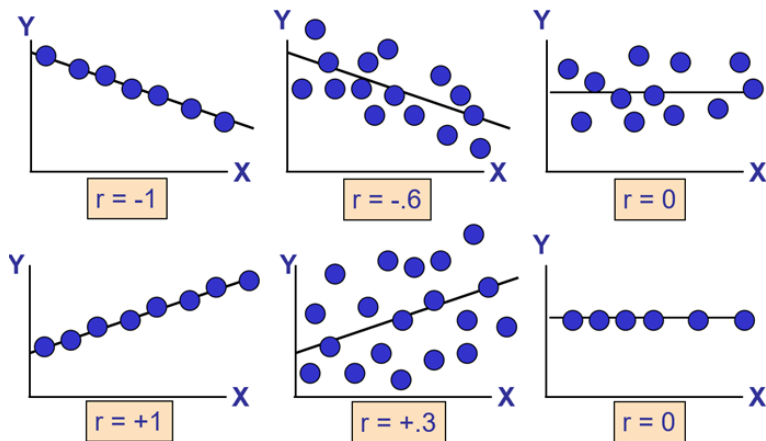
$$-1 \leq r \leq 1$$

$$r = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x)}\sqrt{\text{Var}(y)}}$$

- Measures the relative strength of the **linear** relationship between two variables
- Unit-less: ranges between -1 and 1
- The closer to -1, the stronger the negative linear relationship
- The closer to 1, the stronger the positive linear relationship
- The closer to 0, the weaker any positive linear relationship

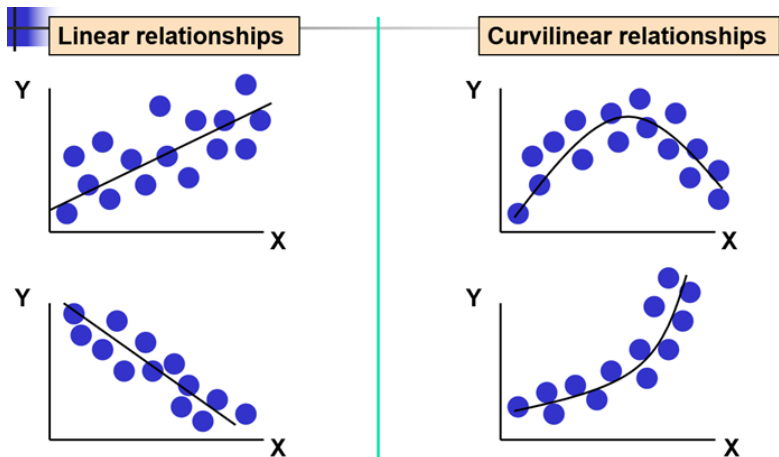
* Rank correlation: spearman's rank correlation.

Introduction: Correlation Coefficient



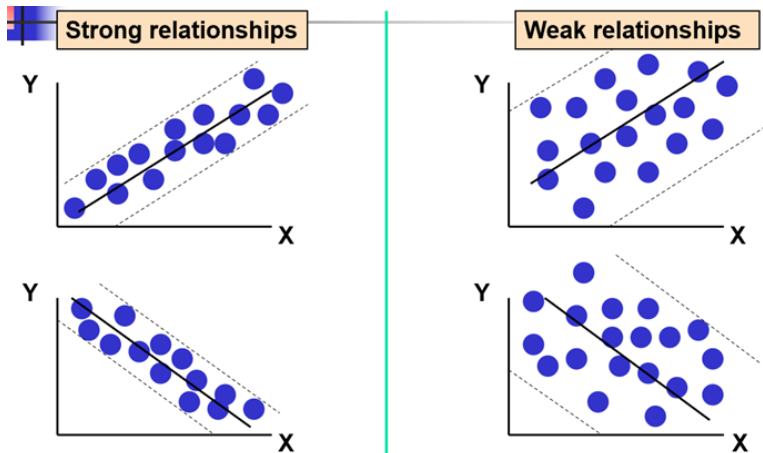
Scatter Plots of Data with Various Correlation Coefficients

Introduction: Correlation Coefficient

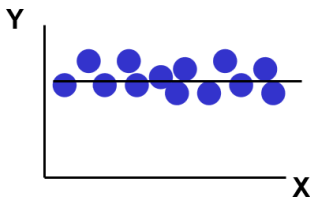
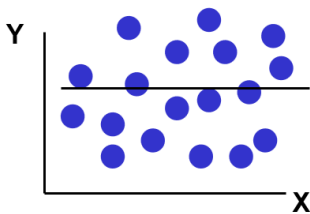


Linear relationships and Curvilinear relationships

Introduction: Linear Correlation



Introduction: Correlation Coefficient



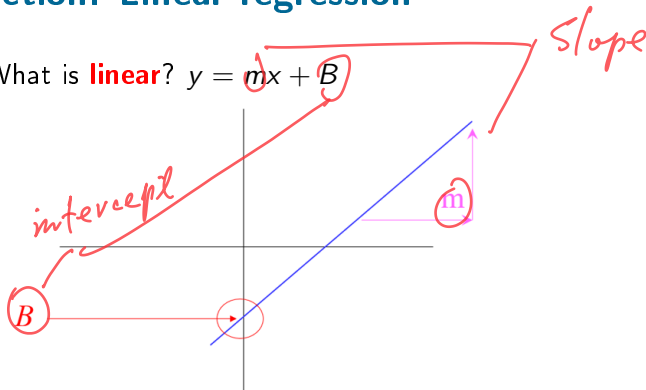
No relationship

Introduction: Linear regression

- In correlation, the two variables are treated as equals.
- In regression,
 - one variable is considered **independent** (=predictor) variable (x) and
 - the other **dependent** (=outcome) variable y .

Introduction: Linear regression

- What is **linear**? $y = mx + B$



- What is the Slope m ?
- A slope of $m = 2$ means that every 1-unit change in X yields a 2-unit change in Y .

Introduction: regression equation and prediction

Data set $\{(y_i, x_i), i=1, \dots, n\}$

■ Prediction

- If you know something about x , this knowledge helps you predict something about y .
- Sound familiar? sound like conditional probabilities?

■ Regression equation

- Expected value of y at a given level of x

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \epsilon_i \sim N(0, \sigma^2)$$

- $\beta_0 + \beta_1 x_i$ is fixed – exactly on the line;
- ϵ_i is an random error, following a (normal) distribution.

* $y_i | x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2) \leftarrow \underline{\text{model}}$

* How to estimate $\beta_0, \beta_1, \sigma^2$

* How good is the model?

Assumptions: regression equation

Linear regression assumes that

- The relationship between x and y is linear;
- y is distributed normally at each value of x ;
- The variance of y at every value of x is the same (homogeneity of variances);
- The observations are independent.

Linear Regression – a brief history

- The earliest form of regression was the method of **least squares**, which was published by **Legendre** in 1805, and by **Gauss** in 1809.
- Legendre and Gauss both applied the method to the problem of determining, from astronomical observations, the orbits of bodies about the Sun.
- Gauss published a further development of the theory of least squares in 1821, including a version of the **Gauss–Markov** theorem.

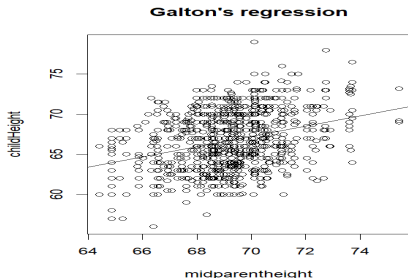


Linear Regression – a brief history

- The term **regression** was coined by **Francis Galton** in the 19th century to describe a biological phenomenon.
- The phenomenon was that the heights of descendants of tall ancestors tend to regress down towards a normal average (a phenomenon also known as regression toward the mean)



Galton's Data



```
library("HistData")
```

```
xx = GaltonFamilies $ midparentHeight
```

```
yy = GaltonFamilies $ childHeight
```

```
plot(xx,yy,xlab="midparentheight", ylab="childHeight",  
main="Galton's regression")
```

```
abline(lm(yy~xx))
```

```
summary(lm(yy~xx))
```

*(father + 1.08 * mother) / 2*

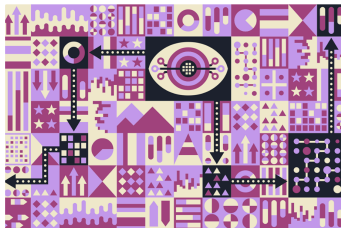
lm(yy ~ xx)

AI and Statistics 2025

AI and statistics

Artificial intelligence Statistics (mathematics)

Delve into the intersection of the dynamic fields of generative AI and statistics to uncover new methodologies and applications.



Mike West

Arts & Sciences Distinguished Professor Emeritus of Statistics & Decision Sciences, Duke University

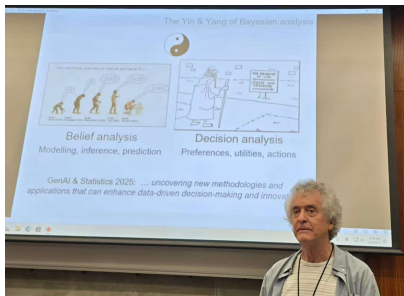
Mike West earned a BSc in Mathematics in 1978 and a PhD in Mathematics (Statistics) in 1982 from the University of Nottingham. He worked at Warwick University before joining Duke University in 1988, where he served as the Director of the Institute of Statistics and Decision Sciences (ISDS) from 1990 to 2001.

As an English and American statistician, Mike's primary focus is on Bayesian statistics. His research contributions span theory and application in various fields, including finance, commerce, macroeconomics, climatology, engineering, genomics, and other areas of biology. His current core research and development activities include Bayesian statistics and decision analysis in causal inference, dynamic modeling, forecasting and time series analysis, dynamic networks, model uncertainty, and simulation and optimization. Since 1990, he has held the position of Arts & Sciences Distinguished Professor of Statistics and Decision Sciences in the Department of Statistical Science at Duke University.



<https://projects.croucher.org.hk/advanced-study-institutes/generative-ai-and-statistics>

AI and Statistics 2025



Formalisms: Some history, some questions

$$p(\mathbf{y}_t | \mathcal{H}_t)? \quad \mathcal{H}_t = h_{t,1:J}(\cdot)$$

- Bayesian model averaging
- Linear pooling: Mixtures
- "Stacking" methods
- ...

- Model-specific calibration, biases, ...
- Incomplete model set
- Domains of expertise: Multivariate
- Model inter-dependencies
- Predictive & decision goals?
- Foundations / theory?

(Bayesian) econometrics: Macro-economics/Central bank forecasting:
 [Fawcett et al '13; Aastveit et al '14; Kapetanios et al '15; Amisano & Geweke '17; Aastveit, Van Dijk & many others 20-24;
 MW + Nakajima, McAlinn, Aastveit et al: '13-20;]

<https://projects.croucher.org.hk/advanced-study-institutes/generative-ai-and-statistics>